25901230 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT201 - 3D PRINTING AND ITS APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Recall the powder bed fusion process using a laser. [U] / [CO1]

   a). Material Jetting

   b). Binder Jetting

   c). SLS

   d). Sheet Lamination

2. What is a major advantage of parametric assemblies in CAD systems? [U] / [CO2]

   a). They allow automatic updates of related components when one dimension changes

   b). They eliminate slicing requirements

   c). They reduce extrusion temperature

   d). They prevent post-processing needs

3. Orientation of parts in build preparation software mainly affects [An] / [CO3]

a). Printer color

b). File storage size

c). Build time and mechanical strength

d). User interface

4. Warping primarily occurs due to [C] / [CO4]

a). Excess extrusion

b). Uneven cooling and shrinkage

c). Low resolution

d). Incorrect slicing

5. Classify CNC machining under manufacturing types. [U] / [CO1]

a). Additive process

b). Subtractive process

c). Forming process

d). Sheet bonding

6. State the goal of computational design. [U] / [CO2]

a). Automated performance based optimization

b). Manual modeling only

c). Hand sketching

d). Removing digital workflow

7. Customization in 3D printing is most beneficial for [An] / [CO3]

a). Large batch identical products

b). Complex and low-volume parts

c). Simple shapes only

d). High tooling-based processes

8. Preventative maintenance reduces [C] / [CO4]

a). Downtime b). Material options

c). Print quality d). Resolution

9. Recognize an advantage of AM in aerospace sector. [U] / [CO1]

a). High material waste

b). Limited geometry

c). Complex lightweight structures

d). Heavy tooling cost

10. Name the software stage that prepares a model for printing. [U] / [CO2]

a). Rendering b). Slicing

c). Simulation d). Polishing

11. While framing AM applications, which cost element is significantly reduced? [An] / [CO3]

a). Material cost

b). Energy cost

c). Tooling cost

d). Transportation cost

12. The controller board in a 3D printer kit is responsible for [C] / [CO4]

a). Heating the nozzle

b). Mechanical stability

c). Holding filament

d). Interpreting G-code and controlling movements

13. Classify post processing under AM stages. [U] / [CO1]

a). Pre production stage

b). Production stage

c). Post production stage

d). Design stage

14. Name the analysis used to evaluate stress in a model. [U] / [CO2]

a). Thermal coloring b). Finite Element Analysis

c). Manual drafting d). Rendering

15. The concept of printing spare parts only when required is known as [An] / [CO3]

a). On-demand manufacturing b). Batch manufacturin

c). Mass customization d). Lean inventory

16. Which representation divides the model into small volumetric elements for AM? [An] / [CO3]

a). Surface representation

b). Boundary representation

c). Voxel representation

d). Wireframe representation

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Discuss the role of energy input and scan speed in controlling microstructure during Directed Energy Deposition. [U] / [CO1]

18. Define topology optimization. [U] / [CO2]

19. Distinguish between industrial and consumer applications in manufacturing. [An] / [CO3]

20. Explain the advantages of Binder Jetting. [U] / [CO1]

21. Explain the advantages of extrusion based printing. [U] / [CO1]

22. Explain the importance of digital workflow integration. [U] / [CO2]

23. Indicate the factors affecting the choice of tooling method. [An] / [CO3]

24. Comprehend the issue of layer shifting and describe its impact on print quality. [C] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25.	Explain the advantages of powder based printing.	[U] / [CO1]
26.	Explain the importance of assembly modeling.	[U] / [CO2]
27.	Relate material properties to functional performance in AM components.	[An] / [CO3]
28.	Design a 3D printed component while justifying choices related to wall thickness,infill density,and support structures.	[C] / [CO4]

[12x2 = 24]